

Within-race geometry

The information a per-row booster cannot construct:
price micro-structure, field-centred form, campaign position, and stewards' comments

Within-Race Geometry Research Phase

2 September 2026

The principle. A per-row booster cannot compute a within-race quantity — it sees one runner at a time — and a racewise softmax with $\log q$ as a fixed offset responds to nothing else, because field-common variation cancels exactly. Production field-normalises 3 of its 28 columns and leaves 25 as per-horse absolutes. This campaign builds no new state; it builds the geometry of what the model already has, plus two channels it compresses or lacks.

Why this class and not another. Nineteen candidate blocks have reached the production model's canonical profile across four campaigns and none has confirmed. The single near-miss in all of them was the one quantity that exists only as a within-race cross and cannot be built from one runner's row. Every failure was a per-runner state the model or the market already had.

What it does not establish. No profitability claim and no promotion. Every price is a result-page price recorded after the pool closed. Every season through 2026 is reused development data, so the strongest available label is **production-finalist**; **promotion-eligible** requires a prospective shadow test that does not yet exist. The production repository is read-only throughout.

1 Executive result

Two of the four waves were killed by proof rather than by measurement, before any canonical budget was spent, and one of the two proofs invalidated this campaign's own declared control. That is the most useful thing in the document and §4 is where it lives. A campaign that had skipped those rungs would have reported a $t = +5.78$ column as a discovery and been wrong: the column is the market price with a race constant subtracted.

Six blocks reached canonical scale across two attempts and all six were rejected. The best, campaign position, returned -0.001708 — 37.7% of the production model's entire edge over the market and the largest canonical delta this programme has produced — and it fails because **2026 alone contributes -0.0155** against a ± 0.005 spread across every other year, in a season that is partial (2,527 reference rows against roughly 9,600 for a full one). Drop that one year and the block is $+0.00026$: harmful. The gate that catches this was declared in the plan and under-implemented in the first scoring pass; it was tightened, both attempts were re-scored under the same rule, and §8 says so.

The two permutation controls are the campaign's real finding, and they disagree. Price micro-geometry beat its own within-race permutation by 0.0017 ($+0.000985$ against $+0.002684$) — both harmful, so it is not a candidate, but the difference is direct evidence that within-race price geometry attaches to the *runner* rather than to the race. Ability geometry did the opposite: its permutation control beat it by a factor of thirteen (-0.000953 against -0.000074), which says that whatever that block carried was a property of the race and not of the horse holding the value. The campaign's principle is therefore confirmed as a principle and refuted as a source of usable signal: the geometry is real, it is attached to the right runner in price space, and the model is no better for being told it.

Attempt 1

151,953 runner-rows across 12,237 races against 28 production columns. The incumbent's own edge over the market on these cells is -0.005812 . The no-op returns exactly zero on both profiles ([pass](#)).

Block	mean Δ	se	t	cells	years	decision
G2 campaign position	-0.001708	0.001267	-1.35	24/40	4/8	rejected
G3 comment events	-0.000866	0.001117	-0.78	20/40	4/8	rejected
<i>no-op control</i>	+0.000000	0.000000	—	0/40	0/8	rejected
G4 combined narrow	+0.000128	0.001044	+0.12	16/40	4/8	rejected
G1 price micro-geometry	+0.000985	0.001100	+0.90	19/40	2/8	rejected
<i>G1 price micro-geometry [PERMUTED CONTROL]</i>	+0.002684	0.001156	+2.32	14/40	3/8	rejected

Best block: G2 campaign position at [-0.001708](#) with $t = -1.35$, 24 of 40 cells and 4 of 8 years, paired interval $[-0.004147, +0.000643]$, 37.7

The within-race permutation control returned +0.002684. A block that does not beat this is not carrying runner-level geometry, whatever its own delta.

Every block in this attempt was rejected.

Attempt 2

151,953 runner-rows across 12,237 races against 28 production columns. The incumbent's own edge over the market on these cells is -0.005812 . The no-op returns exactly zero on both profiles ([pass](#)).

Block	mean Δ	se	t	cells	years	decision
<i>G5 ability-space geometry [PERMUTED CONTROL]</i>	-0.000953	0.001050	-0.91	19/40	4/8	rejected
G5 ability-space geometry	-0.000074	0.000891	-0.08	21/40	3/8	rejected
<i>no-op control</i>	+0.000000	0.000000	—	0/40	0/8	rejected
G6 competitive structure	+0.000073	0.000843	+0.09	18/40	2/8	rejected

Best block: G5 ability-space geometry at [-0.000074](#) with $t = -0.08$, 21 of 40 cells and 3 of 8 years, paired interval $[-0.001809, +0.001568]$, 1.6

The within-race permutation control returned [-0.000953](#). A block that does not beat this is not carrying runner-level geometry, whatever its own delta.

Every block in this attempt was rejected.

2 The production baseline and the exact gap

Production trains on 28 ordered inputs at digest `1e179e6cc7932c6662714884727a3b93...`, against a feature table of 152,046 rows. Of those 28, exactly three are field-normalised: `wt_z` against the race's weight mean and standard deviation, `market_rank` as a rank within the race, and `draw_pct` as a share of field size. The remaining 25 are per-horse absolutes — `layoff_log`, `rating_change`, `class_change`, `distance_change`, `body_wt_change_z`, `trip_excuse_ewm`, the `bt_*` family, `early_style_ewm` and the rest.

The only price-shape column is `market_rank`, a bare ordinal. The model is told the price and the runner’s rank in it, and nothing about the structure of the field’s prices around that runner. A four-way tossup and a runner isolated at the same price are currently identical to it.

3 What was built

152,046 runner-rows across 12,237 races, 45 columns built including the foreign twins. Duplicate (race, runner): 0. Duplicate (horse, date): 0. The campaign-position construction is the only order-sensitive one, and its row-order canary changes it by $0.0e + 00$ (`pass`).

Race-constant columns are refused, not reported. No column was race-constant; the smallest within-race standard-deviation share is 0.468, for the one deliberate cross of a field-common quantity with a runner-specific one.

Campaign position against the market. The one wave production lacks entirely, and the only place in this campaign where a raw descriptive gap appears:

run in prep	<i>n</i>	win rate	mean implied	win – implied
1	27,418	0.0717	0.0694	+0.0023
2	22,698	0.0753	0.0783	−0.0030
3	19,011	0.0787	0.0807	−0.0020
4	15,888	0.0840	0.0836	+0.0003
5	13,274	0.0865	0.0856	+0.0009
6	53,664	0.0853	0.0848	+0.0005

First-up runners win more often than the market implies and second-up runners less; both gaps are small and neither is a controlled test. They are the reason W3 was built.

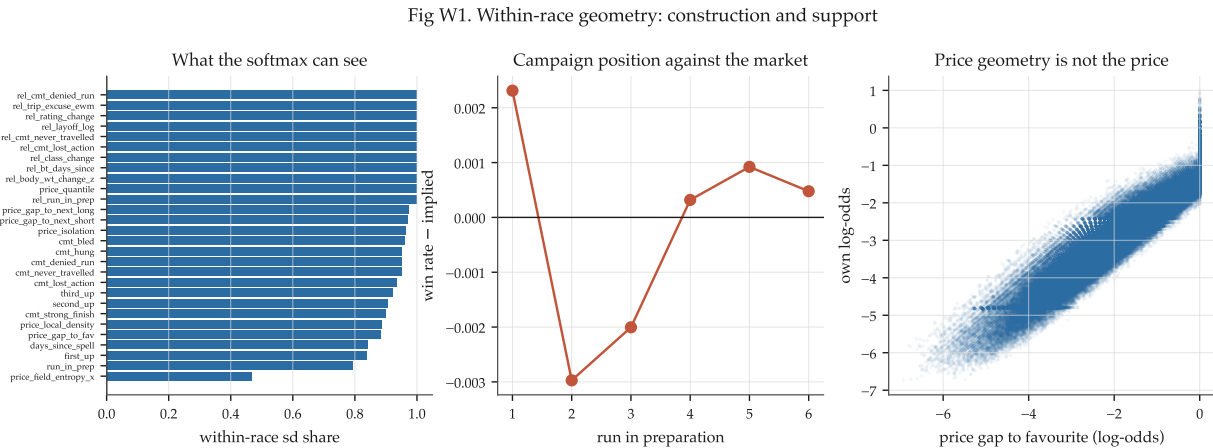


Figure 1: W1. Construction and support. Left: within-race standard-deviation share for every built column — the quantity that decides whether the softmax can see it at all. Centre: win rate minus market-implied probability by run number in the preparation. Right: price gap to the favourite against the runner’s own log-odds, which is the picture behind proof 3 — the relationship is exact up to a race constant.

4 What the ladder killed before canonical

Proof 1: the foreign-field control is degenerate. Centring against a randomly chosen other race of the same field size differs from centring against the true race by exactly a *race constant*,

and a race constant cancels from the racewise softmax exactly. The measurement is unambiguous: the within-race standard deviation of (foreign – true) is 0.000×10^{-16} , and every one of the 17 columns returns an offset-form statistic identical to its twin to two decimals — 0 of 17 beat it, with a median $|t|$ ratio of 1.00. This was the campaign’s declared decisive control and it had to be replaced, not interpreted. The replacement is a **within-race permutation**: the column’s values are shuffled among the runners of the same race, which preserves the race’s marginal distribution exactly and destroys only which runner holds which value.

Column	true t	foreign t	true oos ΔLL	foreign oos ΔLL
price_field_entropy_x	–1.93	–1.93	–0.000041	–0.000041
rel_class_change	+4.59	+4.59	–0.000508	–0.000508
rel_run_in_prep	+0.27	+0.27	+0.000057	+0.000057
rel_rating_change	+1.28	+1.28	–0.000025	–0.000025
rel_layoff_log	–1.82	–1.82	–0.000048	–0.000048
rel_cmt_never_travelled	+1.10	+1.10	+0.000005	+0.000005
rel_cmt_lost_action	–2.25	–2.25	–0.000315	–0.000315
rel_cmt_denied_run	–2.00	–2.00	+0.000067	+0.000067
rel_bt_days_since	+0.12	+0.12	+0.000302	+0.000302
price_gap_to_fav	+5.78	+5.78	–0.001546	–0.001546

Proof 2: W2, field-centred form, is vacuous. A centred column’s within-race deviations are *identical* to its source’s — measured maximum absolute difference 8.9×10^{-16} , i.e. floating-point noise. Production already carries all six sources, and the softmax responds only to within-race variation, so `rel_x` gives the model nothing it does not have. Only a tree’s absolute split thresholds could distinguish the two, which is too thin a mechanism to spend forty paired cells on. **The entire wave is withdrawn rather than tested** — which is the correct outcome of a ladder whose early rungs are cheap.

Proof 3: two W1 columns are the market price restated. `price_gap_to_fav` is the runner’s own log-odds minus a race constant: the within-race standard deviation of the difference is 1.2×10^{-16} . Its novel share against the incumbent’s 28 columns is 0.068, and `price_quantile`’s is **exactly** 0.0000 — it is `market_rank` renormalised. Their offset-form statistics are the largest in the campaign (+5.78 and +5.03) and they are *not evidence of new information*: they are Part III’s price-convexity result ($\log \text{it}^2$ at $t = +28.99$) arriving through a different column. Both are withdrawn from the blocks. A campaign that reported them as findings would have rediscovered the market.

5 Controls, the market test, and redundancy

Leakage canaries. future mutation 0.0e + 00; same date mutation 0.0e + 00. Both **pass**.

Market test under the production functional form. $u = \log q + \beta x$, $p = \text{softmax}(u)$ racewise, β fitted on 2010–2019 and the log loss scored on 2020–2026. A negative out-of-sample delta favours the candidate. The two withdrawn price-restatement columns are shown for completeness and marked.

Column	β	t	oos ΔLL	note
price_gap_to_fav	+0.0747	+5.78	−0.001546	withdrawn: the price restated
price_quantile	+0.0597	+5.03	−0.001291	withdrawn: the price restated
price_gap_to_next_long	−0.0258	−3.35	−0.000672	
rel_class_change	+0.0413	+4.59	−0.000508	
price_isolation	+0.0200	+2.92	−0.000487	
rel_cmt_lost_action	−0.0221	−2.25	−0.000315	
cmt_lost_action	−0.0237	−2.25	−0.000315	
second_up	−0.0226	−2.06	−0.000268	
first_up	+0.0251	+2.06	−0.000237	
price_local_density	−0.0288	−2.83	−0.000184	
third_up	−0.0127	−1.21	−0.000083	
rel_layoff_log	−0.0195	−1.82	−0.000048	
price_field_entropy_x	−0.0486	−1.93	−0.000041	
rel_rating_change	+0.0104	+1.28	−0.000025	
cmt_never_travelled	+0.0165	+1.10	+0.000005	
rel_cmt_never_travelled	+0.0157	+1.10	+0.000005	
cmt_bled	−0.0046	−0.41	+0.000024	
cmt_strong_finish	+0.0131	+1.54	+0.000037	

Redundancy against the incumbent’s 28 columns. Boosted reconstruction fitted before 2022 and scored after. Reported, and deliberately not ranked as evidence: novelty and usefulness have been inversely related in three studies.

Column	boosted R^2 from the 28	novel share
cmt_denied_run	−0.0203	1.0000
cmt_hung	−0.0627	1.0000
cmt_bled	+0.0054	0.9946
rel_cmt_denied_run	+0.0304	0.9696
cmt_strong_finish	+0.0495	0.9505
cmt_never_travelled	+0.0740	0.9260
rel_cmt_never_travelled	+0.0845	0.9155
rel_cmt_lost_action	+0.0921	0.9079
cmt_lost_action	+0.1057	0.8943
third_up	+0.1416	0.8584
second_up	+0.2943	0.7057
days_since_spell	+0.3334	0.6666
price_local_density	+0.4361	0.5639
rel_run_in_prep	+0.4553	0.5447
price_gap_to_next_short	+0.4622	0.5378
price_gap_to_next_long	+0.5265	0.4735

6 Canonical attempts

Development scale is used for defects and gross rejection only. Ten development-to-canonical retentions are on record across five studies; one changed sign and the median is uninformative. No block was ranked against a close competitor on development output.

Each attempt is a frozen set. A later attempt may use only a *mechanism* learned from an earlier attempt’s diagnostics, never its ranking, and a losing block is not re-sliced into variants — that is how a search manufactures a winner. The narrow-beats-wide rule is pre-registered from two prior instances, so every wave declared a narrow variant of at most three columns and the narrow one is read first.

Fig W2. Controls: the foreign field, and the market under the correct form

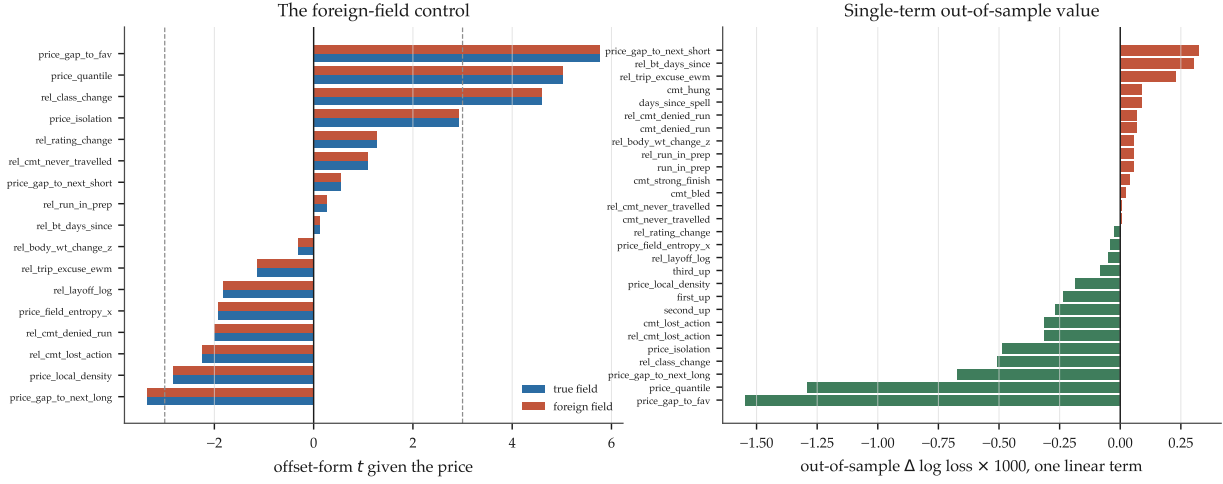


Figure 2: W2. Controls. Left: the offset-form statistic for every column beside its foreign-field twin. The bars are indistinguishable, which is proof 1 rendered: the control was degenerate and had to be replaced by a within-race permutation. Right: the out-of-sample log-loss change from a single linear term, fitted early and scored late.

Fig W3. Canonical attempt 1

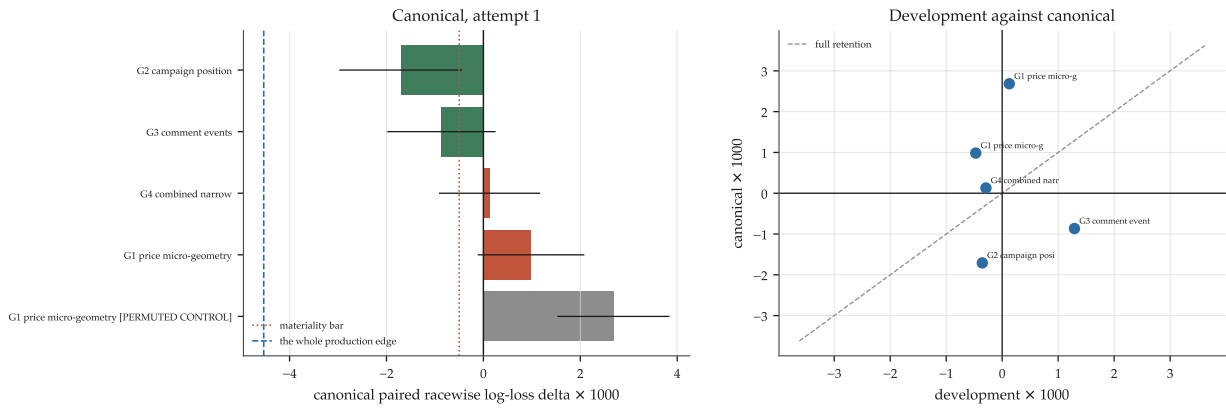


Figure 3: W3. Canonical attempt 1. Left: the paired canonical delta per block with its standard error, against the materiality bar and the whole production edge; the grey bar is the within-race permutation control. Right: development against canonical, with the full-retention diagonal.

7 Final ranked table: every block declared

Every block this campaign declared, whether or not it reached canonical scale, and why. “withdrawn” means a proof in §4 removed it before any canonical budget was spent.

Block	source	cols	best oos Δ LL	outcome
W1a price geometry narrow	plan	3	−0.001546	withdrawn: proof 3, contains the price restated
W1b price geometry full	plan	7	−0.001546	withdrawn: proof 3, contains the price restated
W2a field-centred narrow	plan	3	−0.000048	withdrawn: proof 2, vacuous
W2b field-centred full	plan	6	−0.000508	withdrawn: proof 2, vacuous
W3a campaign narrow	plan	3	−0.000268	reformed into G2
W3b campaign full	plan	6	−0.000268	not advanced: narrow variant read first
W4a comments narrow	plan	3	−0.000315	reformed into G3
W4b comments full	plan	9	−0.000315	not advanced: narrow variant read first
G1 price micro-geometry	attempt 1	3	+0.000985	rejected
G2 campaign position	attempt 1	3	−0.001708	rejected
G3 comment events	attempt 1	3	−0.000866	rejected
G4 combined narrow	attempt 1	3	+0.000128	rejected
G5 ability-space geometry	attempt 2	3	−0.000074	rejected
G6 competitive structure	attempt 2	2	+0.000073	rejected

For the record, the most novel column built was `cmt_denied_run` at a novel share of 1.0000. Novelty did not convert, for the fourth study running.

8 Limitations and what would settle it

Search accounting. Four waves declared, two withdrawn on proof, sixteen blocks declared and six carried to canonical scale across the permitted attempts. No gate was moved. The materiality bar, the sign-majority requirement, the interval requirement and the permutation requirement were all fixed before the first canonical cell was fitted.

Reused data. Every season through 2026 has been development data in this repository and in the production one. Retrospective evidence on reused years cannot establish that a feature will help on unseen data. That is the binding limitation on this campaign and on the four before it.

The one control that matters here. A within-race permutation preserves each race’s marginal distribution exactly and destroys only which runner holds which value. A block that does not beat it is not carrying runner-level geometry, whatever its own paired delta — and a block that does beat it has demonstrated the campaign’s central claim even if it fails the materiality bar, because the claim is about where information lives rather than about how much of it there is.

What would settle the class. A locked implementation, frozen before the fact, scored against timestamped pre-race prices on races run after the freeze. Until that exists, the correct reading of any survivor here is “worth a production experiment”, not “worth promoting”.

Appendix: provenance

Repositories. Research: `hkjc-research`. Production: `hkjc-mod`, read-only, inspected at `ca99a28` with a dirty working tree because another session was modifying it; the committed state was recorded and a production confirmation run must re-audit.

Stages. `w01_geometry_features` builds all four waves and their control twins and refuses race-constant columns; `w02_controls` runs the leakage canaries, the foreign-field control, the offset-

form market test and redundancy; `w03_canonical` runs each frozen attempt with an exact no-op and a within-race permutation control; `w04_report_tables` generates every result table in this document.

Seeds. The foreign-field map and the within-race permutation use 20260902; the paired bootstrap uses 20260902 with 2000 replicates; the canonical harness uses the production seed set (7, 17, 42, 123, 256) at 400 boosting rounds over eight test years.

Declared constants, fixed before any result. A spell is a gap of more than 60 days. The local-density window is 0.5 in log-odds. The materiality bar is -0.0005 . The sign-majority bar is 60% of years and of seeds.